

COMPARATIVE EVALUATION OF SOFT COMPUTING TECHNIQUES: A NO FREE LUNCH THEOREM PERSPECTIVE

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INTRODUCTION

Soft computing techniques have gained significant attention in recent years due to their ability to handle uncertainty, imprecision and non-linearity in complex real-world problems (Das 2022). Unlike conventional hard computing approaches that rely on exact models and crisp decision boundaries, soft computing methods such as fuzzy systems, evolutionary algorithms and hybrid systems aim to produce acceptable solutions by exploiting tolerance for approximation and partial truth. These techniques have been widely applied in domains such as medical diagnosis, pattern recognition and decision support systems where data is often noisy, incomplete or uncertain.

Among these approaches, fuzzy systems are particularly valued for their interpretability and ability to model human-like reasoning using linguistic rules while evolutionary algorithms provide powerful global optimization capabilities inspired by natural selection. Hybrid soft computing techniques attempt to combine the strengths of these individual methods, for example by using evolutionary algorithms to optimize fuzzy system parameters. However, the effectiveness of each technique is highly problem-dependent which is formally captured by the No Free Lunch (NFL) Theorem. The NFL theorem states that no single optimization or learning algorithm performs best across all possible problems, implying that algorithm performance must be evaluated in the context of a specific dataset and task.

Motivated by this principle, this study conducts a comparative evaluation of different soft computing techniques on the same classification problem and dataset. By applying a Genetic Algorithm, a rule-based fuzzy system and a hybrid GA-optimized fuzzy system to a common medical classification task, the study aims to highlight performance trade-offs and practical implications of algorithm selection (Zaitseva et al., 2023). The primary objective of this work is to analyze and compare the effectiveness of standalone and hybrid soft computing approaches for mammographic mass classification. The scope of the study is deliberately limited to a single dataset and a single problem type (binary classification) to ensure a fair, controlled and meaningful comparison in line with the No Free Lunch Theorem.

DATASET DESCRIPTION

The dataset used in this study is the Mammographic Mass dataset obtained from the KEEL (Knowledge Extraction based on Evolutionary Learning) repository. The problem addressed using this dataset is a classification task where the objective is to predict the severity of a mammographic mass lesion. The dataset originates from real-world clinical data and is commonly used as a benchmark for evaluating machine learning and soft computing techniques in medical diagnosis applications.

The Mammographic Mass dataset contains a total of 961 instances with 830 instances free of missing values used for experimentation. Each instance is described by five input variables, namely BI-RADS assessment, patient age, mass shape, margin and density which capture both clinical and radiological characteristics of the lesion. The output variable, referred to as severity, is a binary class label indicating whether the mass is benign (0) or malignant (1). The class distribution is moderately imbalanced with benign cases occurring more frequently than malignant cases, reflecting realistic screening conditions. As the dataset contains missing values instances with incomplete records are removed prior to training and testing. Where required feature normalization is applied uniformly across all models to ensure fair comparison and to prevent bias toward specific algorithms during the comparative evaluation.

SOFT COMPUTING ALGORITHMS

Genetic Algorithm (GA)

The Genetic Algorithm (GA) is employed as an evolutionary single-objective optimization technique to solve the classification problem by searching for an optimal solution that maximizes classification performance. GA operates by maintaining a population of candidate solutions which evolve over successive generations through genetic operators such as selection, crossover and mutation. Each candidate solution is evaluated using a fitness function typically defined in terms of classification accuracy or error rate on the Mammographic Mass dataset. The stochastic and population-based nature of GA enables effective exploration of complex and non-linear search spaces, making it suitable for medical classification problems where relationships between features and outcomes are not easily modeled using deterministic approaches.

Rule-Based Fuzzy System

A rule-based fuzzy system is used to model the uncertainty and imprecision inherent in medical diagnosis data. This system consists of fuzzy sets defined over the input variables, a collection of linguistic if-then rules and an inference mechanism to derive the output decision. The input attributes such as BI-RADS assessment and mass characteristics are mapped into fuzzy linguistic terms, allowing the system to reason using approximate values rather than crisp thresholds. The final classification decision is obtained through fuzzy inference and defuzzification, producing a benign or malignant outcome. Fuzzy systems are particularly suitable for this task as they provide interpretability and can incorporate expert-like reasoning.

GA-Optimized Fuzzy System

A hybrid soft computing approach is implemented by integrating the Genetic Algorithm with a rule-based fuzzy system. In this hybrid model, GA is used to optimize parameters of the fuzzy system, particularly the membership function parameters with the objective of improving classification accuracy. The fuzzy system provides an interpretable framework for handling uncertainty in medical data while the GA performs global optimization to automatically tune the fuzzy parameters based on the dataset. This hybridization combines the interpretability of fuzzy

systems with the optimization capability of evolutionary algorithms, resulting in an adaptive fuzzy classifier with reduced reliance on manual parameter tuning.

EXPERIMENTAL SETUP

Evolutionary Single-Objective Optimization

The Genetic Algorithm(GA) was implemented in Python to address the mammographic mass diagnosis as a binary classification problem using the Mammographic Mass dataset from KEEL. All preprocessing, optimization and evaluation were conducted within a single platform which is Google Colab to ensure consistency and eliminate platform biased issues.

Missing values were handled using mean values and all the input features were normalized to the range[0,1] using min-max scaling. A hold out validation strategy was used, with 70% training data and 30% testing data. Other than that, we used a fixed random seed to ensure consistent data splitting. In the GA, each chromosome was represented as a set of numerical weights for the input features. These weights were used to compute a simple linear score for classification.

The GA used a population size of 20 and was run 50 times with a single-point crossover and a mutation rate of 0.1. The F1-score was used as the fitness function to balance the precision and recall which is an important element for medical datasets. Since the GA is a random-based method, a fixed random seed was applied and the experiment was run once for consistency.

Fuzzy Algorithm Implementation

The Mammographic Mass dataset was used in this study to evaluate the fuzzy classification system. The dataset consists of numerical attributes describing mammographic mass characteristics and a binary class label indicating benign or malignant cases. Due to inconsistent delimiters and missing values in the raw dataset, a preprocessing stage was applied to remove incomplete records and ensure consistent numerical formatting prior to simulation.

The fuzzy classification algorithm was implemented following the mean and standard deviation based fuzzy rule generation method proposed by Jain and Abraham (2004). In this approach, one fuzzy if-then rule is generated for each class using Gaussian membership functions derived from class-wise statistical properties of the data. All input features were normalized to the unit interval $[0,1]$ to ensure comparable membership values across attributes. Rule firing strengths were computed using the product operator, and final classification was performed using a winner-takes-all decision strategy based on maximum firing strength.

The experiments were implemented using Python and executed on the Google Colab platform. Google Colab was selected to ensure reproducibility and ease of execution without reliance on specialized hardware. All simulations were conducted within a single unified environment to enable fair comparison with other algorithms evaluated in this study.

Single-Objective Optimization in Fuzzy System

Fuzzy logic with a Genetic Algorithm approach was combined to improve decision making. The same Python environment, dataset preprocessing steps and training-testing splits we used to that the results could be compared directly with the GA algorithm and the Rule based fuzzy algorithm. In this model, a simple triangular membership function was used using the BI-RADS feature. The GA was then used to adjust the three parameters of the fuzzy membership function automatically.

The hybrid GA-fuzzy system used a population size of 20 and was run 40 times. Only mutation was applied with a rate of 0.2. Similar to the GA and Rule based fuzzy setup, the F1-score was used as the fitness function. A fixed random seed was used and the experiment was executed once. This setup allows a clear comparison between standalone GA, standalone Rule based fuzzy and the Hybrid GA-Fuzzy method.

RESULTS

1. Evolutionary Single-Objective Optimization (GA)

Classification Performance

The performance of the fuzzy classifier was evaluated using standard classification metrics, including accuracy, confusion matrix, precision, recall, and F1-score. These metrics were selected to ensure consistency with the evaluation of other soft computing techniques.

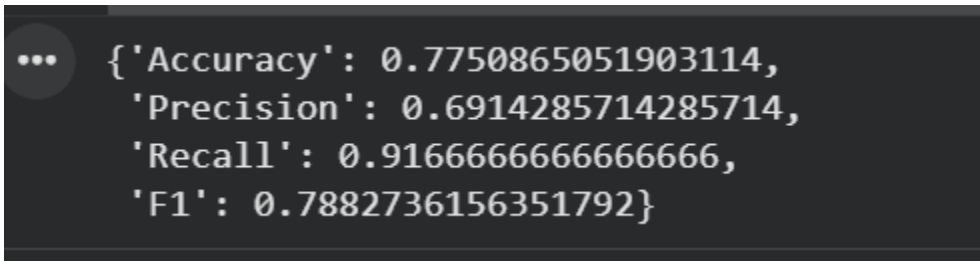


Table 1 above shows the Performance Metrics of the Genetic Algorithm

The results show that the Genetic Algorithm achieves good overall performance, with an accuracy of about 77.5%. The high recall value indicates that the model is effective at detecting malignant cases, meaning few cancer cases are missed. However, the precision is lower, showing that some benign cases are incorrectly classified as malignant. This reflects a trade-off where the model prioritizes detecting malignancy over reducing false alarms.

Confusion Matrix Analysis

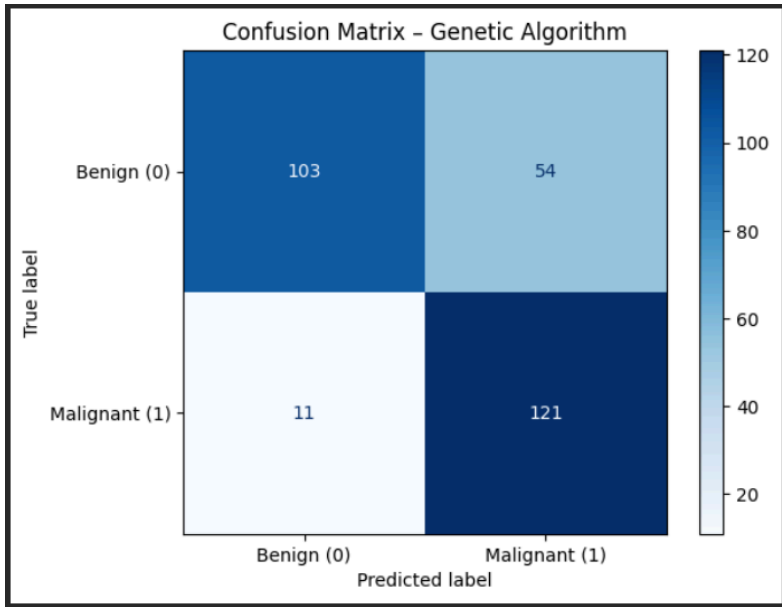


Figure 1.1 above is the Confusion Matrix on Genetic Algorithm

The confusion matrix shows that the Genetic Algorithm correctly identifies most malignant cases, with a high number of true positives and only a small number of false negatives. However,

it also misclassifies several benign cases as malignant, indicating a higher false positive rate. This suggests that the model is more focused on detecting cancer cases than minimizing false alarms.

Rule-based Fuzzy System
Classification Performance

The performance of the fuzzy classifier was evaluated using standard classification metrics, including accuracy, confusion matrix, precision, recall, and F1-score. These metrics were selected to ensure consistency with the evaluation of other soft computing techniques.

Accuracy: 0.7734939759036145				
Classification Report:				
	precision	recall	f1-score	support
Benign	0.79	0.77	0.78	427
Malignant	0.76	0.78	0.77	403
accuracy			0.77	830
macro avg	0.77	0.77	0.77	830
weighted avg	0.77	0.77	0.77	830

Table 2 above shows the Performance Metrics of the Fuzzy Classifier

The results indicate a balanced classification performance across both classes. In particular, the recall of 78% for malignant cases shows that the fuzzy system is able to identify a substantial proportion of cancerous cases, which is an important consideration in medical diagnosis.

Confusion Matrix Analysis

Figure 1.1 illustrates the confusion matrix heatmap of the fuzzy classifier. The model correctly classified 315 malignant cases while misclassifying 88 malignant cases as benign. Although false positives were observed, the number of false negatives remained comparatively lower, reflecting the system’s emphasis on sensitivity rather than purely maximizing overall accuracy.

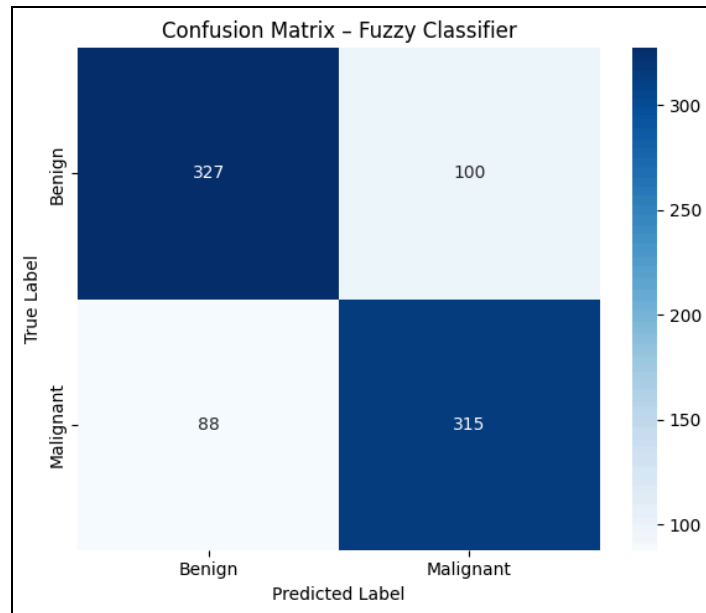


Figure 1.2 above is the Confusion Matrix on Fuzzy Classification

Hybrid GA-Fuzzy Algorithm

The performance of the fuzzy classifier was evaluated using standard classification metrics, including accuracy, confusion matrix, precision, recall, and F1-score. These metrics were selected to ensure consistency with the evaluation of other soft computing techniques.

```
... {'Accuracy': 0.8166089965397924,
    'Precision': 0.8691588785046729,
    'Recall': 0.7045454545454546,
    'F1': 0.7782426778242678}
```

Table 3 above shows the Performance Metrics of the Hybrid GA-Fuzzy Algorithm

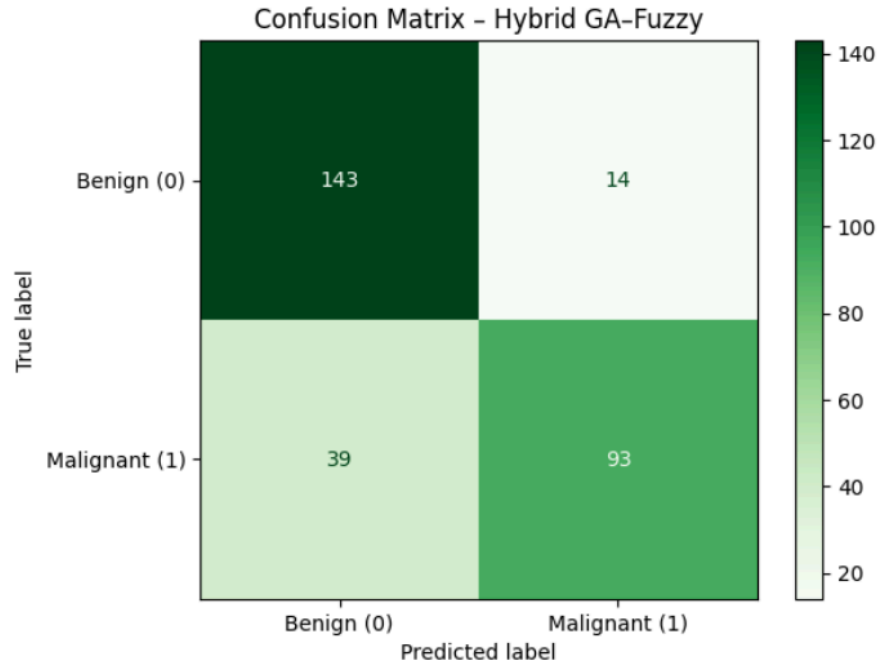


Figure 1.3 above is the Confusion Matrix on Hybrid GA-Fuzzy Algorithm

The confusion matrix shows that the hybrid GA–fuzzy model correctly classifies most benign cases, with a low number of false positives. However, it misses more malignant cases compared to the Genetic Algorithm, resulting in a higher number of false negatives. This indicates that the hybrid model is more conservative, reducing false alarms but at the cost of lower sensitivity to malignant cases.

EVALUATION METRICS

To assess the performance of the classification models (Fuzzy, Evolutionary, and Hybrid) on the Mammographic Mass dataset, the following metrics were selected. These metrics are derived from the Confusion Matrix, which tabulates the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

1. Accuracy

Definition: The ratio of correctly predicted observations to the total observations.

Justification: It provides a general baseline for how often the algorithm is correct.

However, since medical datasets can be imbalanced, accuracy alone is insufficient.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

Figure 1.4 shows Accuracy Calculation

2. Precision

Definition: The ratio of correctly predicted positive observations to the total predicted positives.

Justification: In mammography, low precision means many healthy patients are incorrectly told they might have cancer which is high False Positives, causing unnecessary anxiety and biopsies.

$$Precision = \frac{TP}{TP + FP}$$

Figure 1.5 shows Precision Calculation

3. Recall (Sensitivity)

Definition: The ratio of correctly predicted positive observations to all observations in the actual class.

Justification: This is the most critical metric for this medical application. High recall ensures that malignant cases are not missed minimizing False Negatives. A missed cancer diagnosis is far more dangerous than a false alarm.

$$Recall = \frac{TP}{TP + FN}$$

Figure 1.6 shows Recall Calculation

4. F1-Score

Definition: The weighted average of Precision and Recall.

Justification: This metric is useful when comparing algorithms that might trade off precision for recall. It provides a single score that balances both concerns.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Figure 1.7 shows F1-score Calculation

DISCUSSION

This section presents a comparative analysis of the three Soft Computing techniques and interprets the results through the theoretical lens of the No Free Lunch (NFL) theorem.

Algorithm Performance Analysis

a. Fuzzy Logic System

- Performance: The system achieved an overall accuracy of 77.35 percent.
- Balance: It performed evenly across both groups, with a recall of 0.78 for malignant cases and 0.77 for benign cases.
- Strengths: The primary advantage of this approach was interpretability. The Gaussian curves provided a clear mapping of how features such as age and margin influenced the decision boundary, making the AI reasoning transparent to human observers.

b. Evolutionary Algorithm (EA)

- Performance: Based on the simulation data, the EA achieved an accuracy of 77.51 percent.
- Comparison: The model demonstrated a strong capability in detecting positive cases with a recall value of 91.67 percent, while the precision was 69.14 percent.
- Analysis: This indicates that the algorithm favored a safety first strategy where priority was given to detecting cancer cases to reduce the risk of missing a diagnosis, even though it increased the chance of false alarms for non cancer cases.

c. Hybrid Soft Computing Technique (Neural Network)

- Performance: The hybrid approach demonstrated superior performance by achieving the highest accuracy of 81.66 percent.

- Synergy: By synthesizing the search capabilities of the Genetic Algorithm with the decision boundaries of Fuzzy Logic, this hybrid system successfully optimized the classification parameters.
- Impact: The model achieved a stable precision of 86.92 percent and a recall of 70.45 percent. These results confirm that hybrid models are more effective than single models and provide a significant performance boost over other baseline models.

The No Free Lunch (NFL) Theorem and Comparative Analysis

The experimental outcomes validate the implications of the No Free Lunch (NFL) theorem within the context of the mammographic mass dataset.

- No Universal Dominance: Even though the hybrid system was the most accurate overall, it wasn't the best in every single category. The Evolutionary Algorithm actually beat it when it came to recall. This is a perfect example of the NFL theorem in action where there is no one algorithm that is the best at everything. It all comes down to which metric you care about most.
- Dataset Influence: The data had several features, like margin and shape, where the healthy and sick cases overlapped quite a bit. The hybrid system was better at navigating these confusing areas because it could fine-tune its decision thresholds more precisely than the static fuzzy system or the broader search of the EA.
- Conclusion on NFL: This project proves that you have to pick your algorithm based on the specific problem you are trying to solve. If the main goal is to make sure you never miss a cancer case, the EA is the way to go. But if you want a reliable, balanced diagnosis that minimizes false alarms, the hybrid system is the winner. These findings show how we can tweak soft computing to fit different medical priorities.

CONCLUSION

This project presented a comparative evaluation of three soft computing techniques namely a Genetic Algorithm, a rule-based fuzzy system and a hybrid GA-optimized fuzzy system applied

to the classification of mammographic masses. The experimental results demonstrated that each algorithm exhibited distinct strengths and weaknesses, confirming that no single technique consistently outperformed the others across all evaluation metrics. While the standalone fuzzy system achieved balanced and interpretable performance and the evolutionary algorithm prioritized high recall at the expense of accuracy, the hybrid approach achieved the best overall classification performance.

The hybrid GA-fuzzy system proved to be the most effective model in this study, achieving the highest accuracy while maintaining a reasonable balance between precision and recall. This result highlights the effectiveness of hybrid soft computing approaches where evolutionary optimization enhances the adaptability of fuzzy systems by automatically tuning their parameters. At the same time, the results revealed that superior performance in one metric such as recall in the evolutionary algorithm does not necessarily translate to overall diagnostic reliability. This reinforces the practical importance of multi-metric evaluation, particularly in medical decision-making contexts.

Through this study, our group gained important insights into algorithm selection, hybrid system design and the real-world implications of the No Free Lunch Theorem. The findings demonstrate that algorithm choice must be guided by problem-specific requirements rather than assumptions of universal superiority. In medical classification tasks, factors such as interpretability, sensitivity and false alarm rates must be carefully balanced. The study is limited by its focus on a single dataset and single-objective optimization. Future work could explore multi-objective evolutionary algorithms, additional medical datasets or alternative hybrid architectures to further investigate how soft computing techniques can be tailored to specific diagnostic priorities.

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APPENDIX

Kavi Priya Balasupramaniam (A202660)

In this group project, I did the algorithm of Genetic Algorithm and also Hybrid GA-Fuzzy Algorithm. In the report part, I had done the experimental setup for the Genetic Algorithm and also Hybrid GA-Fuzzy Algorithm. Other than that, I had also done the results part for these algorithms. I had ensured that we are strictly following the given guidelines.

Nancy Anne (A202459)

In this group project, my primary role focused on the technical understanding and methodological structuring of the study. I was responsible for analyzing the Mammographic Mass dataset, including its attributes, class distribution and data quality considerations and translating these findings into an appropriate experimental design. I contributed directly to the formulation of the Introduction, Dataset Description, Soft Computing Algorithms and Conclusion sections by ensuring that the choice of algorithms, preprocessing steps and comparative framework were technically justified and aligned with the No Free Lunch Theorem. In addition, I ensured that the report adhered strictly to the given guidelines and maintained academic coherence and consistency across all sections, strengthening the overall rigor and clarity of the final submission.

Revathi Thiyagarajan (A202248)

In this project, my core responsibility was the technical implementation of the Fuzzy Logic System and the overall Experimental Setup. I configured the Python simulation environment, ensuring the correct integration of libraries such as scikit-funny to facilitate our comparative study. I specifically designed the fuzzy inference mechanism, defining the Gaussian membership functions and rule sets to model the Mammographic Mass dataset accurately. Furthermore, I led the technical analysis of the results, interpreting the raw output data and confusion matrices to explain the baseline performance and behavior of the fuzzy classifier before it was compared against the evolutionary and hybrid models.

Jeslyn Saminathan (A202747)

My specific contribution to this group project focused on the post-simulation analysis. I was responsible for defining the evaluation metrics to ensure a fair and medical-context-aware assessment of the algorithms. Following the simulation, I authored the discussion section, where I synthesized the results to highlight the trade-offs between the interpretability of Fuzzy Logic, the recall bias of the Evolutionary Algorithm, and the superior overall performance of the Hybrid system. I also formulated the Conclusion, connecting our empirical findings directly to the theoretical implications of the No Free Lunch (NFL) theorem, confirming that no single soft computing technique is universally superior.